

Phoenix public parks serve as tree refuges in deprived areas

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1 Introduction

Urban trees, whether in private yards, planted along streets, or in public parks, are a critical component of sustainable cities. The urban forest provides numerous ecosystem services that benefit city residents, including shade, stormwater removal, and air pollution mitigation ([Fulton et al, 2025](#)). Tree canopy cover (TCC) — defined as the area of ground covered by tree leaves, trunks, and branches — is especially crucial in hot arid cities like Phoenix, Arizona, where ambient temperatures in full sun can feel up to 12.6 °C hotter than under tree shade ([Colter et al, 2019](#)). Adequate shade can be a lifesaving amenity in Phoenix, which experienced 337 heat deaths in 2024 ([of Public Health, 2024](#)).

Like many urban amenities, tree canopy cover is often inequitably distributed across cities. Several studies have shown that residents of lower socioeconomic status tend to live in neighborhoods with lower tree canopy cover ([Greene et al, 2018](#), [Schwarz et al \(2015\)](#), [Lowry et al \(2012\)](#)). Some studies in U.S. cities also report that neighborhoods with larger proportions of ethnic and racial minorities tend to have less tree canopy ([Schwarz et al, 2015](#), [Locke et al \(2021\)](#), [Foster et al \(2024\)](#), [Elderbrock et al \(2024\)](#), [Walker et al \(2024\)](#), [Zhou and Kim \(2013\)](#)). These patterns have been shown to hold true in Phoenix as well. Median income level has consistently been shown to have a positive correlation with TCC in the city, while the proportion of Hispanic residents

within block groups is negatively correlated with TCC (Wang et al, 2024, Nesbitt et al (2019), Nelson et al (2021)). However, the proportion of Black residents was negatively correlated with TCC in one study (Wang et al, 2024) , while another study found no correlation between these variables (Nesbitt et al, 2019).

As a source of publicly available tree canopy cover, municipal parks could potentially act as “tree refuges” and compensate for the lack of tree cover in disadvantaged neighborhoods. While park equity studies have yielded mixed results, some have shown consistent patterns of disparities in park access in the United States. A 2016 literature review by Rigolon et al. found that neighborhoods with high White populations had the greatest park access in 18 out of 20 papers examined (Rigolon, 2016).

The research on disparities in access to tree canopy cover in parks is limited. A spatial analysis of urban parks across the U.S. found that parks are more common in areas with larger White populations, and these parks tend to have greater tree coverage (Winkler et al, 2024). Fossa, et al. (2023) investigated parks within the three largest U.S. cities and found that park tree cover was positively associated with larger Black populations, larger Hispanic populations, and more socioeconomic deprivation in surrounding neighborhoods (Fossa, 2023). Conversely, Elderbrock, et al. (2024) found that park tree canopy cover is equitably distributed across Portland, Oregon, regardless of the racial, ethnic, or economic makeup of the surrounding neighborhoods (Elderbrock et al, 2024).

Several researchers have analyzed disparities in Phoenix park access. Lara-Valencia (2018) found that Hispanic residents of Phoenix tend to have greater access to municipal parks due to a higher concentration of parks in the urban core (Lara-Valencia and Garcia-Perez, 2018). However, Ibes (2015) analyzed access to public parks in Phoenix by park type and found that “green neighborhood parks” are more concentrated in neighborhoods with higher White populations while “urban core parks” are more often found in areas with lower incomes and more minority residents (Ibes, 2015). Corley (2024) found no association between park acreage and minority resident proportion or poverty level at the block group level in Phoenix (Corley, 2024). However, these three studies on Phoenix park access did not account for spatial autocorrelation of variables—the tendency of spatially proximate observations to be more similar than expected under random distribution—which can inflate the apparent significance of predictors when left unaddressed (Anselin, 1995). Kim, et al. (2023) addressed this limitation by using spatial regression models to investigate park access in Phoenix and nearby Mesa, Arizona, and found that residents of more affluent neighborhoods had access to greater park acreage within walking distance, but there were no significant correlations between race/ethnicity and the number of parks, size of parks, or distance to the nearest park (Kim et al, 2023).

As seen in the Phoenix studies cited above, research on park access tends to use a simple Euclidean distance metric to determine accessibility, but this method does not account for the actual distance that residents must travel to visit parks. The travel distance from a residential area to a public amenity can vary greatly from the Euclidean distance depending on the layout of the road network (Li et al, 2019).

Additionally, most studies consider “park access” to be a binary variable, where a resident has complete access to any parks within a given catchment area and no access to parks outside of the catchment area. However, this approach does not capture the reality of park visitation—people are more likely to visit nearby parks, but may still travel a greater distance to visit larger parks. For example, Kim et al. (2025) found that Phoenix residents tend to favor local parks within 1 km of their neighborhood, but the likelihood that they will travel beyond 1 km to visit larger amenity-rich parks increases as the size of the local parks decreases (Kim et al, 2025).

Traditional park access research also tends to only address the “supply” side of access by quantifying park number, area, or amenities (Kim et al, 2025, Nesbitt et al (2019), Winkler et al (2024)) while ignoring the “demand” side, or the number of people competing for access to the parks. Other studies incorporate the population factor by simply dividing the total park area by the population within a given catchment (Lara-Valencia and Garcia-Perez, 2018). The Gaussian two-step floating catchment area (G2SFCA) method of assessing park access addresses these shortcomings by calculating distance to an amenity based on road network rather than Euclidean distance and incorporating a park supply factor (e.g., total area or amenities), a demand factor (the population of residents with access to the parks), and a travel distance metric that weights closer parks more heavily than distant parks (Zhang et al, 2023a).

Residential tree canopy cover provided by trees in neighborhoods and commercial areas likely has a more direct impact on residents’ lives than the tree canopy in public parks since residential trees shade people’s homes, workplaces, and travel routes. But parks may act as tree canopy refuges, allowing people to temporarily enjoy the benefits of trees that may be lacking in their daily lives. No studies have directly investigated the relationship between residential tree canopy cover and access to park tree canopy cover, but a 2013 study by Zhou and Kim investigated neighborhood tree canopy disparities and walking-distance access to parks in six Illinois cities (Zhou and Kim, 2013). They found that in four of the cities, a higher proportion of Black residents was associated with lower neighborhood tree canopy cover, but there were no disparities in access to parks. However, Zhou and Kim did not quantify the tree canopy cover within the parks.

Given the inequities in tree canopy distribution and apparent lack of inequities in park access within Phoenix, it is possible that public parks serve as tree canopy refuges that partially compensate for lack of tree cover in neighborhoods with high minority populations and low socioeconomic status. However, no research to date has tested this hypothesis. This study attempts to fill this research gap by answering the following questions:

RQ1. To what extent do racial/ethnic composition and socioeconomic deprivation predict residential (non-park) tree canopy cover across Phoenix neighborhoods?

RQ2. To what extent do racial/ethnic composition and socioeconomic deprivation predict public park tree canopy accessibility in Phoenix, and does this relationship diverge from the patterns observed for residential tree canopy?

RQ3. Do Phoenix public parks function as tree canopy refuges in areas with low residential tree canopy cover, and does this compensatory pattern hold uniformly across racial/ethnic groups and socioeconomic levels?

This research expands on existing studies of tree canopy cover in Phoenix by isolating residential tree canopy from the canopy cover provided by public park trees to avoid double counting of park trees in the analysis. It also improves on the analyses of park tree canopy access by using the G2SFCA method to capture a more realistic picture of accessibility compared to traditional Euclidean distance methods that ignore population demand and the limited availability of park entrances. And it uses spatially aware regression methods to analyze associations between tree canopy cover and demographic factors to ensure that spatial autocorrelation does not overly influence the models.

2 Materials and methods

2.1 Study area

Phoenix, Arizona is the fifth largest city in the United States, with a 2024 population of about 1.67 million within a 1,342 km² area (Bureau, 2024a). Phoenix has a desert climate with mild winters and extremely hot summers. Daily maximum temperatures frequently exceed 38°C, making Phoenix the hottest large city in the country. Average annual precipitation is 195 mm, with most rainfall occurring during the winter and summer wet seasons (Fan et al, 2015).

Hispanic residents make up 43% of the city's population. The racial makeup of Phoenix is 40% White, 7% Black, 5% Asian, and 1% Native American, with 4% of the population identifying as multiracial. The median household income is \$85,246, with 12% of residents having an income below the federal poverty level of \$15,960 per person (Bureau, 2024a).

2.2 Block groups and parks

Census block groups were used as the unit of analysis in this study because this is the smallest geographic units for which detailed socioeconomic data are available. A block group is an area defined by the U.S. Census Bureau that encompasses several contiguous city blocks and usually contains between 600 and 3000 residents. Block group boundaries as of 2024 were downloaded from the United States Census Bureau using the tidycensus R package (Walker and Herman, 2026).

Phoenix public park boundaries were downloaded from the City of Phoenix Open Data portal (of Phoenix, 2026) and manually edited using QGIS to exclude restricted access areas such as golf courses and botanical gardens. Parks established after 2021 and those missing land cover data were excluded, resulting in 177 public parks in the final dataset. Private planned community parks were excluded from the analysis as they are often inaccessible to non-residents (of Phoenix, ???a).

The City of Phoenix Parks and Recreation Department classifies parks into five types based on their size and amenities. Each park type has a service area radius that

Table 1: Typical amenities, sizes, and service area radius of Phoenix public park types

Park type	Typical amenities
Pocket	Playground; small open area; picnic areas
Linear	Linear open space; walking/biking path; fitness equipment; picnic areas
Neighborhood	Open space; playground; sports fields; fitness equipment; picnic areas
Community	Amenities found in neighborhood parks, plus: swimming pool; recreational center
Regional	Amenities found in community parks, plus: community/senior center; sports complex

represents the recommended distance from a Phoenix residence to the nearest park of a given type (of [Phoenix Parks and Department](#), n.d.). Park type characteristics are summarized in Table 1.

2.3 Tree canopy cover

Tree canopy data were extracted from a 2021 land cover raster file created by the University of Vermont Spatial Analysis Lab and provided by American Forests ([Forests, 2021](#)). The imagery has a spatial resolution of 0.3 m and includes most of Phoenix except for large undeveloped desert preserves and some urban land on the periphery of the city. A total of 1017 block groups had tree canopy cover data available. The percentage of total tree canopy cover (TTCC) per block group was calculated using the zonal histogram tool in QGIS. TTCC represents the total amount of tree canopy cover within a block group, including street trees and those in private yards, businesses, and parks. I then clipped public park polygons from the land cover raster and calculated the percentage of tree cover on the non-public park land within block groups, known as the residential tree canopy cover (RTCC). Figure 1 shows the effect of trees within a public park on the tree canopy cover within a block group. Although RTCC includes trees within non-residential areas such as schools, churches, and parking lots, they are considered “residential” in this study since they are not located in public parks, and they have a more direct impact on residents’ daily lives.

I performed the Mann-Whitney U test to determine whether TTCC within all block groups in the city is affected by the presence of public parks. The median TTCC value of the 176 block groups with parks was compared to the TTCC value of the 841 block groups without parks, and the U statistic was calculated.

I performed the Wilcoxon signed-rank test on the block groups that contain public parks to determine whether the parks contribute to the total tree canopy cover within those block groups.

2.4 Demographic factors

I sourced the total population and race/ethnicity data at the block group level from the 2024 American Community Survey (ACS) 5-year estimate ([Bureau, 2024b](#)) using the tidycensus R package. These factors were used to calculate the percentage of Black and Hispanic residents per block group. Population estimates from the 2024

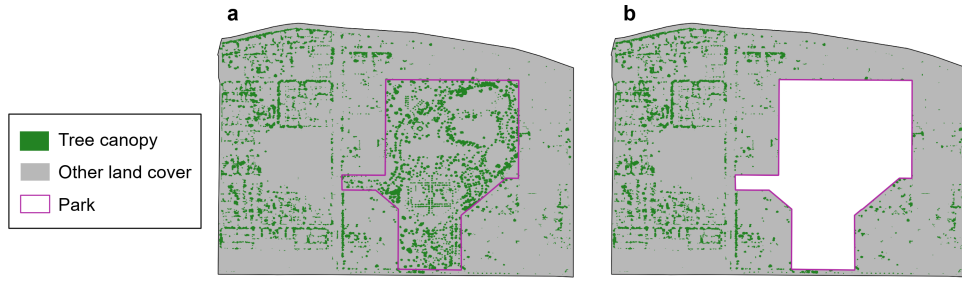


Figure 1: The effect of public park tree canopy cover on total tree canopy cover within a block group: a) Total tree canopy cover (TTCC) including the park = 11.7%, b) Residential tree canopy cover (RTCC) excluding the park = 8.2%

ACS 5-year estimate were also downloaded for individual Census blocks to be used for calculating population-weighted block group centroids.

Research on disparities in tree canopy cover tend to use median income or median household income as an explanatory variable to capture the socioeconomic status of residents [[Lara-Valencia and Garcia-Perez \(2018\)](#), [Zhou and Kim \(2013\)](#)]. Some studies also include variables related to housing and education ([Zhou and Kim, 2013](#)). However, socioeconomic status is multifaceted and not easily summed up by individual factors; therefore, it is common in equity research to use a validated composite measure of social vulnerability to better represent the nuances of economic and social deprivation. One such metric is the area deprivation index (ADI) developed by Singh in 2003 ([Singh, 2003](#)). The ADI incorporates 17 U.S. Census indicators of income, education, employment, and housing quality at the census block group level (600-3000 people). The raw ADI score of a block group is compared against all other block groups in the nation and ranked on a percentile scale from 1-100, where 1 indicates the lowest level of disadvantage and 100 indicates the highest disadvantage. ([Rollings et al, 2023](#)).

I acquired the ADI national ranks of Phoenix block groups from the University of Wisconsin School of Medicine Public Health ([2023](#)). Block groups containing predominantly group housing (college dormitories or jails) do not have ADI data available, so they were excluded from the analysis. The resulting dataset contained 988 block groups used in the demographic analysis. Figure 2 shows the locations of block groups with land cover and/or demographic data available and the types and sizes of public parks within these block groups.

2.5 Spatial access to park tree canopy cover

The Gaussian two-step floating catchment area (G2SFCA) method was used to determine Phoenix residents' spatial access to tree canopy cover in public parks. This method accounts for both the supply of public amenities and demand on those amenities when

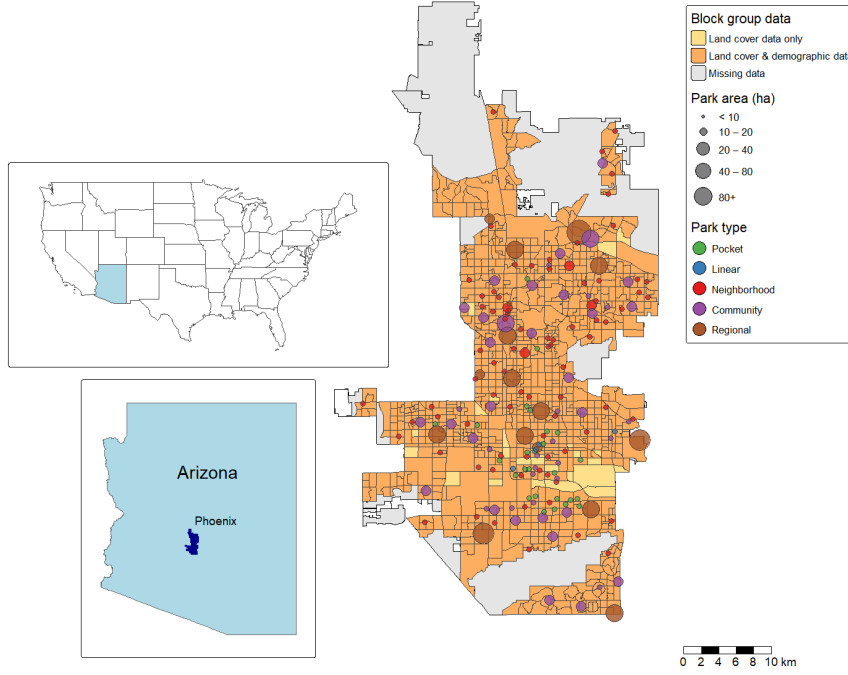


Figure 2: Phoenix block group data availability, park sizes, and park types

Table 2: Total park area and park tree canopy cover area in Phoenix public parks

	Mean	SD	Median	Min	Max
Park area (ha)	8.30	12.51	4.46	0.08	115.25
Park tree canopy cover (%)	15.57	8.24	14.40	0.47	48.99

$N = 177$ parks

calculating accessibility. The population of areas surrounding an amenity represents the demand on the amenity (Hu et al, 2020). In other words, more people living near a park leads to greater competition for the park’s amenities, including tree canopy cover.

Park tree canopy area in hectares (PTCA) was used as the supply term and was calculated by multiplying the percent tree canopy cover of a park by its total area (Table 2). Since PTCA values are affected by both park size and percent tree canopy cover, residents living near several small tree-dense parks may have access to more tree canopy than residents near a single large park with scant tree cover. Figure 3 shows the range of tree canopy area by park type and for all parks combined.

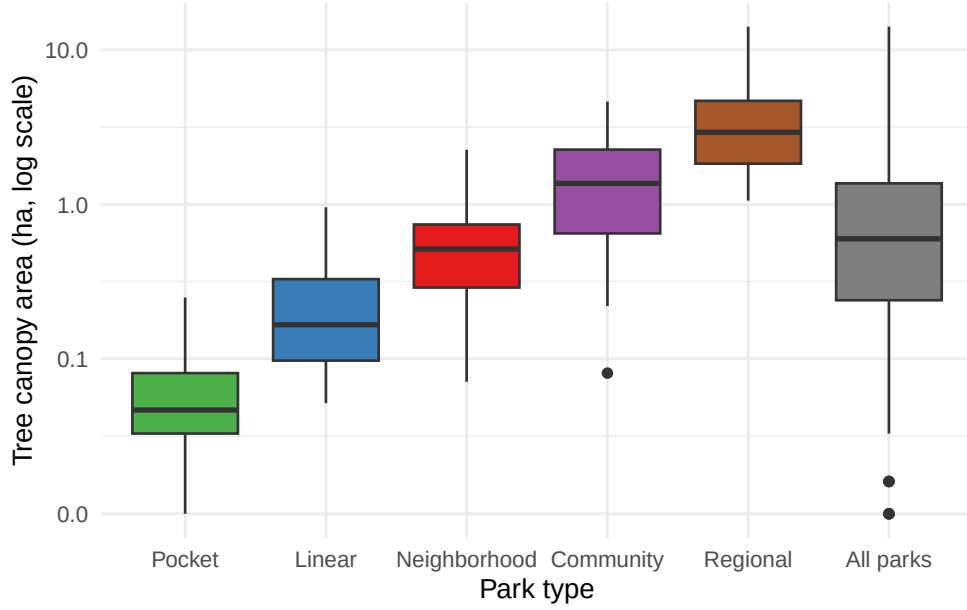


Figure 3: Tree area (ha) by park type

An accurate assessment of park access using the G2SFCA method depends on the choice of supply and demand origin points used to determine travel distance from residents to parks. Census block group centroids are often used as demand origin points in spatial access research because they represent the “average” location of residents within a reasonably small area. However, population-weighted block group centroids are less likely to underestimate accessibility than geographic centroids (Kim and Horner, 2024). For example, the geographic centroid of a large block group with housing concentrated in one corner is likely to be located far from areas where most people live, leading to a suboptimal estimate of the travel distance to a public park.

Likewise, the geographic centroid of a park is commonly used as the supply origin point, but this choice can provide an inaccurate picture of park access, especially in large parks or long linear parks where the centroid is far from the entrances. Residents also may not be able to access parks at any point along the boundary but are instead limited to a small number of entrances. Using park entrances rather than centroids as the supply origin points in a G2SFCA analysis overcomes these limitations and better represents how residents access public parks using various modes of transportation (Zhang et al, 2023a).

The first step in the G2SFCA method is to compute the ratio of park tree canopy area to the surrounding block group population at each park location. The population-weighted centroid of block groups was used as the population location (k) in calculating the distance between block groups and parks. This point represents the mean location

of residents within the block group. Populations at each block group location k are weighted using a Gaussian function. This ratio is calculated as:

$$R_j = \frac{S_j}{\sum_{k \in d_{kj} \leq d_0} G(d_{kj}, d_0) P_k}, \quad (1)$$

where P_k is the population at block group k whose population-weighted centroid lies within the catchment (i.e., $d_{kj} \leq d_0$) of a park at location j ; d_{kj} is the travel distance between block group k and park j ; S_j is the supply capability of park j , i.e., the tree canopy area in hectares; G is the friction-of-distance, calculated below:

$$G(d_{kj}, d_0) = \begin{cases} \frac{e^{-(1/2) \times (d_{kj}/d_0)^2} - e^{-(1/2)}}{1 - e^{-(1/2)}}, & \text{if } d_{kj} \leq d_0 \\ 0, & \text{if } d_{kj} > d_0 \end{cases}. \quad (2)$$

The second step in G2SFCA is to search all parks l within the threshold distance d_0 from block group i , thus creating the catchment for the population at i . Each R is discounted using the Gaussian function G . Any park outside the catchment is assigned a G of 0, assuming residents will not travel long distances to visit parks. The discounted R values within the catchment area of i are summed to obtain the spatial accessibility of park tree canopy area at block group i as follows:

$$A_i = \sum_{l \in d_{il} \leq d_0} G(d_{il}, d_0) R_l, \quad (3)$$

where l denotes all parks within the catchment of block group i , and all other notations are the same as in Equation 1. The park tree canopy accessibility score (A_i) represents the amount of park tree canopy cover (in hectares) available to each resident in a given block group. Raw accessibility scores are very small values and thus were re-scaled by a factor of 1,000,000 for reporting purposes; the re-scaled values are hereafter referred to as park tree access scores (PTAS).

Public park access studies often use park centroids as destination points (Dai, 2011, Hu et al (2020), Peng and Yin (2025)). However, this method can bias the results against large parks since users may have to walk long distances from a park entrance to the centroid (Zhang et al, 2023b). This study used actual park entrances, determined by visual examination in QGIS, as destination locations for parks larger than 2 ha. For smaller parks, the centroid was used since travel distance would be minimally affected by the access point location.

Travel distance (d_{kj}) from block group centroids (k) to park entrances (j) was computed using the driving profile in the osrm R package (Giraud, 2022). This method uses the Open Source Routing Machine to identify the shortest driving path between any two points based on the OpenStreetMap road network. A distance matrix of all block group population-weighted centroids to all park entrances was created and then limited to the distance from a block group to only the nearest entrance within any given park.

Previous studies on park access tend to rely on arbitrary catchment sizes, typically in the range of 0.5 to 1 km, based on the assumption that people will only walk a short distance to visit parks (Elderbrock et al, 2024, Lara-Valencia and Garcia-Perez (2018), So (2016)) . However, Phoenix is a uniquely car-dependent city (Saeidizand and Boussauw, 2024), and Kim et al. (2025) have shown that Phoenix residents are willing to travel farther to access larger parks with more amenities (Kim et al, 2025). Therefore, rather than rely on a single catchment size (d_0) for all parks, d_0 was determined by the service area radius assigned by the City of Phoenix to each park type (pocket, linear, neighborhood, community, or regional) (of Phoenix, n.d.). Pocket and linear parks were assigned a threshold of 0.8 km (0.5 mi), neighborhood parks 1.6 km (1 mi), community parks 4.8 km (3 mi), and regional parks 8.0 km (5 mi). This aligns the park catchment values with the park accessibility goals of the City of Phoenix.

While the focus of this study is on Phoenix residents' access to parks, people from nearby cities are free to visit Phoenix parks, thus diluting the quantity of park tree canopy available to Phoenix residents. Therefore, the populations of 682 block groups from surrounding cities that lie within park catchment areas were also included in the calculation of the supply/demand ratios (R_j and R_l).

2.6 Regression analysis of tree canopy cover disparities

The first step in evaluating demographic disparities in block group-level tree canopy cover was to apply ordinary least squares (OLS) regression models using RTCC and PTAS as the outcome variables and percent Hispanic population, percent Black population, and ADI national percentile rank as the predictor variables. An initial OLS model using the untransformed PTAS score showed non-normal residuals (Shapiro-Wilk $W = 0.843$, $p < .001$) with visible right skew in a Q-Q plot. A square root transformation was applied to PTAS and the model re-estimated; residual normality improved ($W = 0.955$) but remained significantly non-normal ($p < .001$).

Demographic and environmental variables often exhibit spatial autocorrelation, meaning the residuals in a given areal unit are likely to be similar to their surrounding units. The presence of spatial autocorrelation violates the assumption of independent residuals required for OLS regression, necessitating the use of spatially aware regression models (Anselin and Rey, 1991). The global Moran's I index was calculated to measure the overall spatial autocorrelation of the residuals of both OLS regression models using the following formula:

$$I = \frac{N}{S_0} \frac{\sum_{i=1}^N \sum_{j=1}^N w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^N (x_i - \bar{x})^2}, \quad (4)$$

where N is the number of block groups indexed by i and j , x is the variable of interest (i.e., OLS residuals), $\bar{x}$ is the mean of x , w_{ij} is a spatial weights matrix that represents the degree of spatial proximity between block groups i and j . The distance matrix was calculated based on the eight nearest neighbors of each block group. This method allowed for all block groups to be included in the computation, resulting in no isolated

“islands” of block groups. A Monte Carlo simulation with 999 iterations was used, and the p -value was set at 0.01 to evaluate the significance of the global Moran’s I index. The value of Moran’s I ranges from -1 to 1; positive values indicate clustering, negative values indicate dispersion, and 0 indicates a random distribution of residuals.

Since spatial autocorrelation was present in the residuals of the OLS models for both dependent variables, spatial regression models were warranted. Lagrange Multiplier (LM) tests were used to determine the appropriate spatial regression specification for each outcome, following the decision rule recommended by Anselin and Serge ([Anselin and Rey, 1991](#)): if the robust LM error statistic is significant and the robust LM lag statistic is not, a spatial error model is selected; if robust LM lag is significant and robust LM error is not, a spatial lag model is selected. LM tests indicated that a spatial error specification was appropriate for both RTCC and PTAS (see Section 3).

In a spatial error model, spatial dependence is captured in the error term rather than the outcome itself. This specification is appropriate when neighboring block groups have similar residuals due to unmeasured place-based factors that cluster spatially and affect tree canopy cover, such as housing age or homeowners association planting guidelines. By modeling this spatial structure as an error term, the spatial error model isolates the independent effects of racial/ethnic composition and socioeconomic disadvantage on tree canopy outcomes. The spatial error model takes the following form:

$$y = X\beta + u, \quad (5)$$

where y is the outcome (RTCC or PTAS), X represents the predictor variables (percent Hispanic population, percent Black population, and ADI rank), β represents the coefficients, and u is an error term modeled as:

$$u = \lambda Wu + \epsilon, \quad (6)$$

where λ is the spatial error coefficient, Wu is the weighted average of neighboring errors based on a spatial weights matrix of the eight nearest neighbors, and ϵ is a random independent error term.

2.7 Tree canopy access typology

Block groups were classified according to RTCC and PTAS values. Because PTAS already captures park tree canopy as its supply-side variable, using TTCC as an outcome measure would essentially count the same trees twice: once in the block group tree canopy access and once in the park accessibility axis of the typology (see [Fossa, 2023](#)). Use of RTCC avoids the duplication issue.

Block groups with an RTCC value of 15% or higher were classified as high RTCC category, following American Forests’ recommendation that desert cities should maintain a 15% tree canopy cover to achieve tree equity ([of Phoenix, 2022b](#)); the remainder were classified as low RTCC. In the absence of comparable policy guidelines for park tree access, block groups with a PTAS at or above the median were classified as high

PTAS, and those below the median as low PTAS. The two factors were then combined to create four tree canopy access categories: a) high RTCC/high PTAS (HH), b) high RTCC/low PTAS (HL), c) low RTCC/high PTAS (LH), or d) low RTCC/low PTAS (LL).

Multinomial logistic regression was used to examine whether the percent Hispanic population, percent Black population, or ADI rank predicted a block group’s membership in each tree canopy access category relative to the high RTCC/high PTAS condition (HH). Given that spatial autocorrelation was detected in the residuals of both the RTCC and PTAS regression residuals, it was expected that the residuals of the tree access category logistic regression model might exhibit similar spatial dependence. However, global Moran’s I cannot be calculated directly on the residuals of a multinomial logistic model, since the outcome comprises four unordered categories rather than a single continuous or binary residual vector. To address this, each tree canopy access category (HH, HL, LH, LL) was recast as a binary outcome (one-vs-rest) and modeled separately via logistic regression on the same three demographic predictors. Global Moran’s I was then calculated on the residuals of each binary model to test for spatial autocorrelation.

Significant spatial autocorrelation was detected on the residuals of all four binary regression models (see Section 3), so the Moran eigenvector spatial filtering approach was used to incorporate spatial predictors into the models (Griffith and Peres-Neto, 2006). This approach decomposes a spatial weights matrix into orthogonal eigenvectors, each representing a distinct spatial pattern at a different scale. These eigenvectors are incorporated into the regression model as additional predictors, thereby absorbing the spatial signal that would otherwise be part of the model’s error term. Eigenvector selection was implemented using the spatialreg R package (Bivand et al, 2021), using the eight-nearest-neighbors matrix used previously. Each eigenvector was tested iteratively, and those that significantly reduced Moran’s I ($\alpha = 0.08$) were retained. The selected eigenvectors were added as additional covariates to each binary logistic model, and global Moran’s I was recalculated on the resulting residuals to assess the degree to which spatial autocorrelation had been reduced.

3 Results

3.1 Impact of public parks on tree canopy cover

Percent total tree canopy cover (TTCC) is the amount of tree canopy cover in a block group including the tree cover provided by public parks. Across the entire city, the 176 block groups that contain parks have a median TTCC of 10.95, while the 841 block groups without parks have a median TTCC of 11.17. This difference is non-significant (Mann-Whitney $U = 78897$, $p = 0.168$), suggesting that parks are evenly distributed across the canopy spectrum rather than concentrated in already-green neighborhoods.

This comparison does not indicate whether parks add canopy within their own block groups, however. To test this, I compared each park-containing block group’s TTCC to its RTCC directly. Of the 176 block groups that contain parks, the median TTCC was

Table 3: Total tree canopy cover (TTCC) and residential tree canopy cover (RTCC) in block groups that contain parks

	Mean	SD	Median	Min	Max
Total tree canopy cover (%)	11.47	4.19	10.95	3.65	26.28
Residential tree canopy cover (%)	11.25	4.36	10.20	3.49	27.50
TTCC - RTCC (%) [†]	0.23	0.74	0.12	-2.59	3.94

$N = 176$ block groups containing parks

[†]TTCC: Total tree canopy cover; RTCC: Residential tree canopy cover

10.95%, while RTCC was lower at 10.20% (Table 3). This difference was statistically significant (Wilcoxon signed-rank test, $V = 11271$, $p < 0.001$), confirming that parks meaningfully contribute to tree canopy cover when they are present in a neighborhood. Interestingly, although the presence of parks had an overall positive impact on TTCC, 56 (31.8%) of the park-containing block groups saw a reduction in TTCC compared to RTCC. This occurs when a park with low tree canopy cover is located in a block group with a higher canopy cover, thus reducing the neighborhood’s total tree canopy percentage.

3.2 Regression analysis

Table 4 presents descriptive statistics for the block group characteristics and outcome variables (RTCC and PTAS) used in the regression analyses. Block group population is included for reference, but it was not used as a predictor variable. Figure 4 illustrates the RTCC and PTAS values per block group, along with the area of each park covered by tree canopy. Maps of the three predictor variables (percent Hispanic population, percent Black population, and ADI rank) are depicted in Figure 5.

Based on the OLS model, percent Hispanic population, percent Black population, and ADI rank were all negatively associated with RTCC ($p < 0.001$; see Table 5). For every 1% increase in the Hispanic population, residential tree canopy cover within a block group is expected to decrease by 3.5%; a 1% increase in the population of Black residents is associated with a 7.3% decrease in RTCC; and a 1-point increase in the ADI rank is associated with an 8.5% decrease in RTCC.

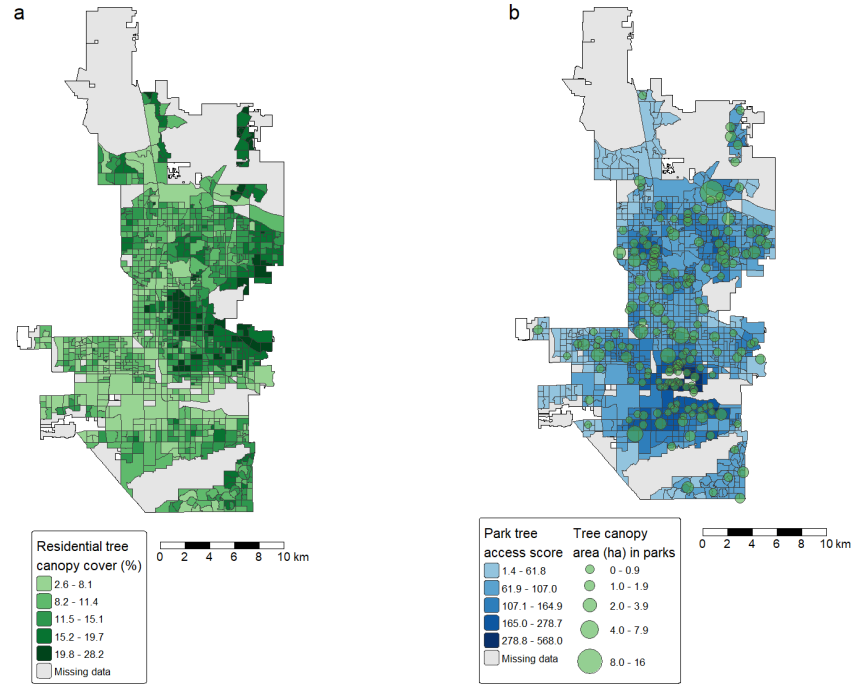
In the OLS model, the percentage of Hispanic residents in a block group is not significantly associated with the square root of PTAS, but there was a significant positive association between the square root of PTAS and both Black resident percentage ($p < 0.05$) and ADI rank ($p < 0.001$; see Table 5). However, with an R^2 of only 0.073, these demographic factors explain very little of the variation in park tree access.

Global Moran’s I was significant ($p < 0.001$) for both the RTCC and PTAS residuals from the OLS models. The RTCC index was 0.221, indicating moderate spatial clustering of residential tree canopy values, and the PTAS index was 0.418, indicating strong spatial clustering of park tree access scores.

Table 4: Descriptive statistics of the dataset

	Mean	SD	Median	Min	Max
<i>Block group characteristics</i>					
Total population	1,597	710	1,499	102	6,221
Hispanic population (%)	39.7	28.7	33.9	0.0	99.3
Black population (%)	6.8	9.5	3.0	0.0	59.1
ADI rank	40	19	40	1	100
<i>Outcome variables</i>					
Residential tree canopy cover (%)	11.9	4.3	11.1	2.6	28.2
Park tree access score	105.7	58.5	97.3	1.4	568.0

$N = 988$ block groups

**Figure 4:** Tree canopy variables: a) Block group residential tree canopy cover (%) and b) block group park tree access score with tree canopy area (ha) within parks

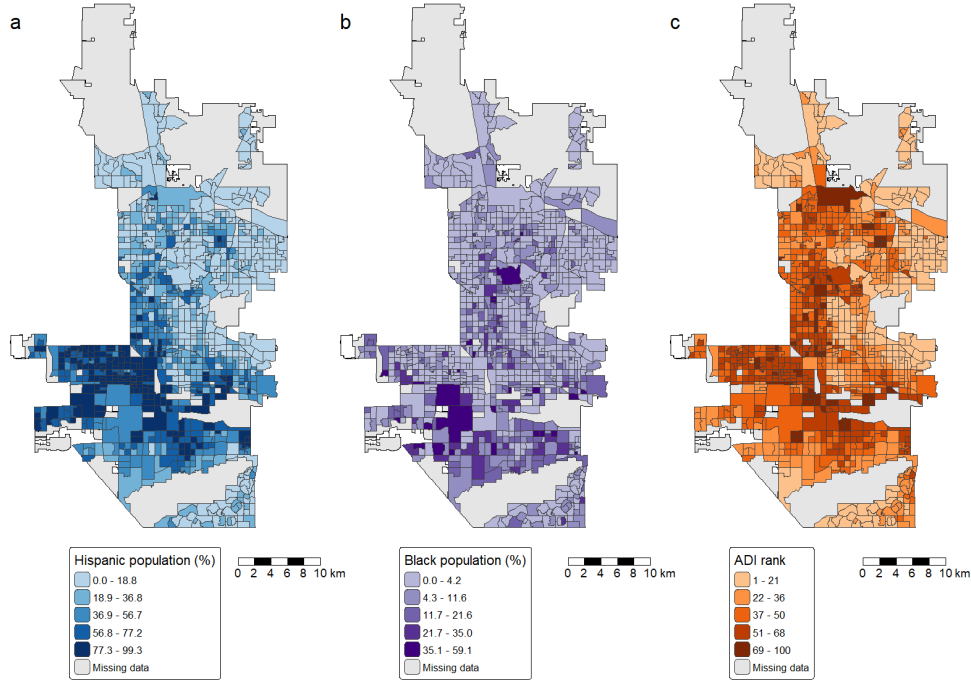


Figure 5: Block group demographic factors: a) Hispanic population (%), b) Black population (%), and c) ADI national percentile rank

Spatial error model results are presented in Table 5 along with OLS estimates for comparison. Reduced Akaike Information Criterion (AIC) values indicate that the spatial error models perform better than OLS models for both RTCC and PTAS (square-root transformed). The lambda (λ) term indicates how neighboring observations influence one another's errors. The positive λ in both spatial error models indicates that unmeasured error in one block group significantly impacts the errors in adjoining block groups for both RTCC and $\sqrt{\text{PTAS}}$. The substantially higher lambda in the $\sqrt{\text{PTAS}}$ models ($\lambda = 0.731$) relative to the RTCC models ($\lambda = 0.462$) likely reflects the spatial smoothing inherent to the G2SFCA construction of PTAS, in which overlapping park catchments induce autocorrelation among neighboring block groups regardless of demographic makeup.

Similar to the OLS model, all three predictor variables are negatively correlated with RTCC in the spatial error model, indicating that block groups with larger Hispanic populations, larger Black populations, or higher ADI rank are more likely to have lower residential tree canopy cover even after spatial autocorrelation is taken into account. The pseudo- R^2 value of 0.417 for the spatial error model fit is consistent with a reasonably strong model fit for RTCC.

Table 5: Results of OLS and spatial error regression models for RTCC and PTAS

	RTCC models ¹		$\sqrt{\text{PTAS}}$ models ¹	
	OLS ²	Spatial error	OLS ²	Spatial error
(Intercept)	17.262***	17.199***	8.402***	8.541***
Hispanic population (%)	-0.035***	-0.040***	0.004	0.000
Black population (%)	-0.073***	-0.069***	0.027**	0.019*
ADI rank	-0.085***	-0.080***	0.030***	0.027***
lambda		0.462***		0.731***
AIC	5251.8	5163.7	4692.0	4291.1
R ²	0.362		0.073	
Pseudo-R ²		0.417		0.383

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

$N = 988$

¹RTCC: Residential tree canopy cover; $\sqrt{\text{PTAS}}$: Park tree access score (square-root transformed)

²R² is reported for OLS models; Nagelkerke pseudo-R² is reported for spatial error models

The pseudo- R^2 value of 0.383 for the square-root transformed PTAS spatial error model is substantially higher than the OLS R^2 (0.073). Given the differing basis of these two statistics, this shift should be interpreted cautiously, but it is consistent with the much larger increase in λ for PTAS relative to RTCC. Both results likely reflect the spatial structure inherent to the overlapping catchment areas in the G2SFCA method. This model is consistent with the OLS results finding no significant association between park tree canopy access and Hispanic population percent. Accounting for spatial structure reduces the percent Black population coefficient in the $\sqrt{\text{PTAS}}$ from 0.027 ($p < 0.01$) in the OLS model to 0.019 ($p < 0.05$) in the spatial error model. This finding is consistent with a portion of the OLS-based association reflecting shared spatial structure in park tree canopy cover rather than a completely demographic effect.

The significant positive relationship between $\sqrt{\text{PTAS}}$ and ADI rank in the spatial error model again indicates that the park tree access score increases as socioeconomic status decreases. This relationship remained stable in both magnitude and significance after accounting for spatial autocorrelation, supporting ADI as the more robust predictor of access to park tree canopy in Phoenix.

3.3 Tree canopy access typology

Block groups were organized into one of four tree canopy access categories based on their RTCC and PTAS values (Table 6). Of the 988 total block groups, 211 met the high-RTCC threshold and 777 fell below the threshold; PTAS values were split evenly around the median ($n = 494$ high and 494 low). The spatial distribution of each

Table 6: Tree canopy access categories

Category	Rule	Description
HH	High RTCC + High PTAS	Abundant residential and park tree canopy
HL	High RTCC + Low PTAS	Abundant residential tree canopy, but low park tree access
LH	Low RTCC + High PTAS	Low residential tree canopy, but parks provide tree canopy
LL	Low RTCC + Low PTAS	Residential and park tree canopy desert

$N = 988$ total block groups

category is illustrated in Figure 6. The least common category is HH, or neighborhoods with both abundant residential tree canopy and higher than average access to park tree canopy. These block groups are mostly confined to isolated pockets in the northeastern part of Phoenix. Block groups in the HL category have abundant residential tree canopy, but lower than average access to park trees. The most common category is LH, representing low-residential canopy neighborhoods where public parks provide a tree canopy refuge to the residents. They are located mostly in the center and northern portions of the city. The remaining block groups belong to the LL category, or tree canopy deserts, lacking both residential tree canopy cover and access to public park tree canopy. They tend to be found on the periphery of the city.

The multinomial logistic regression model estimates the odds that a particular demographic factor (Hispanic population, Black population, or ADI rank) predicts whether a block group belongs to either the HL, LH, or LL tree access category rather than the baseline HH category (Table 7). The results show that block groups are equally likely to be in the HL category compared to the HH category, regardless of the demographic makeup of the block group. In other words, neighborhoods with high residential tree canopy cover and high park tree canopy access are not demographically distinct from those with high residential canopy and low park tree access.

All three demographic factors were significantly associated with membership in the LH category ($p < 0.05$). For every 1% increase in the Hispanic population, there is a 2% increase in the odds that a block group belongs to the LH category vs. the HH category. A 1% increase in Black population is associated with a 5% increase in the odds of belonging to the LH category. A one-point increase in ADI national percentile rank is associated with an 8% increase in the odds of being in the LH category.

For the LL tree access category, only ADI rank is significant ($p < 0.001$). For every one-point increase in ADI rank, there is a 6% increase in the odds that a block group belongs to this category compared to the HH category.

Higher ADI rank was associated with greater odds of belonging to both low RTCC categories, LH and LL, relative to HH. This supports the previous finding that low socioeconomic status is linked to reduced residential tree canopy cover in Phoenix. But there is no clear relationship between the socioeconomic level of a low-RTCC block group and its likelihood of having greater access to public park tree canopy cover.

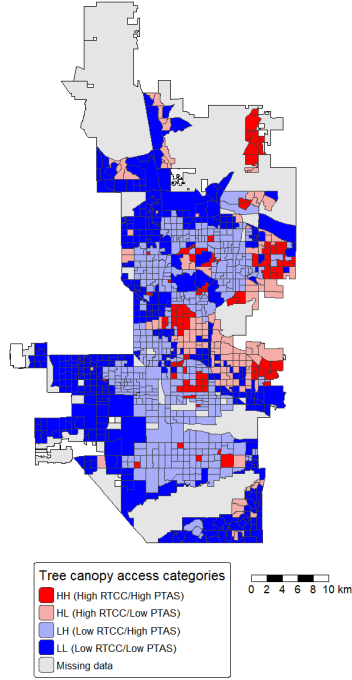


Figure 6: Block group tree canopy access categories

The higher odds of a block group with a larger Black or Hispanic population belonging to the LH category suggests that these neighborhoods are more likely to have parks that act as tree canopy refuges, providing higher park tree canopy access despite having low residential tree canopy cover.

Because spatial autocorrelation was detected in the one-vs-rest binary logistic models underlying the four-category canopy access typology (Moran's I : HH = 0.140, HL = 0.144, LH = 0.267, LL = 0.232; all $p < 0.001$), Moran eigenvector spatial filtering was applied to the models. Selected eigenvectors (HH: 9, HL: 14, LH: 30, LL: 23) were added as covariates to each binary model, substantially reducing spatial autocorrelation across all four categories. Moran's I was reduced by 76-85% in each model (HH = 0.033, HL = 0.030, LH = 0.040, LL = 0.034), though modest statistically significant spatial clustering remained in each case ($p < 0.05$). This result is consistent with spatial structure operating at a scale broader than the local nearest-neighbor weighting could capture.

Table 8 reports the coefficients for each tree canopy access binary logistic models before and after spatial filtering. Both the HH and HL categories were negatively associated with ADI national percentile rank, while the LH and LL categories were positively associated with ADI rank. These associations were significant both before and after spatial filtering, indicating that the relationship between ADI and canopy

Table 7: Multinomial logistic regression odds ratios for canopy access categories

Predictor	HL		LH		LL	
	OR ¹	95% CI	OR	95% CI	OR	95% CI
Hispanic population (%)	1.00	0.98, 1.02	1.02*	1.00, 1.03	1.01	1.00, 1.03
Black population (%)	0.99	0.94, 1.05	1.05*	1.01, 1.10	1.04†	1.00, 1.08
ADI rank	0.99	0.96, 1.01	1.08***	1.05, 1.10	1.06***	1.03, 1.08

Reference category: HH (High RTCC/High PTAS)

† $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

$N = 988$: HH = 89 (9.0%), HL = 122 (12.3%), LH = 404 (40.9%), LL = 373 (37.8%)

¹OR: Odds ratio; CI: Confidence interval

access category is robust to the presence of spatial autocorrelation in the model residuals. Significance levels for most other predictor terms also remained stable after filtering, with two exceptions: for the HH category, the negative associations with Hispanic population and Black population moved from marginal significance ($p < 0.1$) to significance at $p < 0.05$; and for HL, the negative association with Hispanic population shifted from non-significant to significant at $p < 0.05$, while the association with Black population strengthened from $p < 0.05$ to $p < 0.01$.

4 Discussion

This study of distributional justice concerns related to residential tree canopy cover and park tree access in Phoenix, Arizona resulted in three main findings. First, residential tree canopy cover (RTCC) is inequitably distributed: block groups with larger Black and Hispanic populations and higher area deprivation (ADI) have significantly lower RTCC (RQ1). Second, park tree canopy access (PTAS) follows a different pattern: it is not associated with Hispanic population proportion but is significantly *higher* in block groups with larger Black populations and greater deprivation (RQ2). Third, these patterns converge in the overall tree canopy access typology: relative to “canopy abundant” block groups (high RTCC and high PTAS), “tree canopy refuge” block groups (low RTCC but high PTAS) have significantly larger Black and Hispanic populations and higher area deprivation (RQ3).

4.1 Residential tree canopy disparities

Previous studies in North American cities have shown that TTCC is often negatively associated with higher concentrations of racial and ethnic minorities but positively

Table 8: Binary logistic regression results for canopy access categories before and after spatial filtering

Term	(unfiltered)	(filtered)
<i>HH</i>		
(Intercept)	-0.303	-0.778**
Hispanic population (%)	-0.013†	-0.018*
Black population (%)	-0.035†	-0.041*
ADI rank	-0.044***	-0.039***
<i>HL</i>		
(Intercept)	0.577**	0.628*
Hispanic population (%)	-0.01	-0.021**
Black population (%)	-0.043*	-0.066**
ADI rank	-0.065***	-0.068***
<i>LH</i>		
(Intercept)	-2.13***	-2.533***
Hispanic population (%)	0.007*	0.006
Black population (%)	0.019**	0.025**
ADI rank	0.032***	0.038***
<i>LL</i>		
(Intercept)	-0.76***	-1.151***
Hispanic population (%)	0.001	0.005
Black population (%)	0.005	0.001
ADI rank	0.004	0.008

† $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

associated with higher socioeconomic status (Greene et al, 2018, Schwarz et al (2015), Elderbrock et al (2024), Walker et al (2024), Zhou and Kim (2013)), and this pattern has been documented in Phoenix as well (Wang et al, 2024). The current study demonstrates this inequitable pattern of tree canopy cover applies specifically to RTCC, excluding the tree canopy provided by public parks. RTCC is negatively associated with percent Hispanic population and percent Black population, indicating that neighborhoods with a higher concentration of minority residents have lower tree canopy cover provided by trees in private yards, parking lots, schools, and other privately and publicly accessible areas. The negative relationship between RTCC and area deprivation index (ADI) rank is even more pronounced; block groups with more economically vulnerable residents have significantly lower residential tree canopy.

These disparities in tree canopy cover may partially explain why low-income Phoenix communities are disproportionately impacted by urban heat island effects (Zhu et al, 2022). Tree canopy shade can blunt the impact of the urban heat island by reducing radiant temperatures by up to 23.5 °C compared to full sun (Colter et al, 2019). Tree planting efforts in low-income and minority neighborhoods should be a critical component of climate change resilience policies in the Phoenix area to achieve tree canopy equity.

4.2 Park tree canopy access

While tree planting in residential areas should be a major focus of efforts to ameliorate tree canopy disparities in Phoenix, municipal governments have a limited ability to affect the number of trees planted on privately owned land. As freely accessible amenities, public parks can be a useful tool for increasing tree canopy cover in areas with low residential canopy. This study shows that park tree canopy access does not track privilege in the way residential canopy does. In fact, the relationship runs in the opposite direction.

PTAS shows no relationship with the proportion of Hispanic residents within a block group. However, block groups with larger Black populations and greater deprivation have *higher* PTAS than other block groups—the reverse of the pattern found for RTCC. Rather than reproducing existing socioeconomic disparities, PTAS is concentrated in the block groups that plausibly need it the most. One possible explanation is that Phoenix parks are more common in the south and center of the city where Black populations and deprivation are both at their highest. Given the sprawling nature of urban development in Phoenix (Kane et al, 2014), parks near the city center may be older, allowing more time for the tree canopy to develop compared to newer suburban parks. Future work could incorporate park age as a predictor of PTAS to test this hypothesis.

4.3 Parks serve as tree canopy refuges

Given that RTCC is inequitably distributed in Phoenix, while PTAS is higher in socially vulnerable neighborhoods, public parks do seem to serve as tree canopy refuges in areas where this canopy is arguably needed the most. Compared to block groups with abundant RTCC and PTAS, tree canopy deserts (block groups with low RTCC and low PTAS) are much more likely to be socioeconomically disadvantaged and marginally more likely to have large Black populations. However, block groups in which parks serve as tree canopy refuges (those with low RTCC and high PTAS) are also more likely to have large Black populations and high deprivation. This indicates that tree canopy cover within Phoenix public parks is well-situated to at least partially compensate for a lack of residential tree canopy in areas that are most vulnerable to urban heat island effects (Zhu et al, 2022). A concerted effort should be made by the City of Phoenix to increase park tree cover in the 37.8% of block groups that qualify as true tree canopy deserts.

The tree canopy access patterns found here both echo and diverge from Elderbrock et al.'s (2024) study of tree canopy equity in Portland, Oregon. Like this study, the authors found less residential tree canopy in block groups with larger Black, Hispanic, and low-income populations. Unlike this study, however, they found no compensatory pattern in park tree canopy access, which was unrelated to race or income and was actually *lower*, not higher for Hispanic residents (Elderbrock et al, 2024).

This divergence may be due to a difference in function between Portland and Phoenix parks, or it may be methodological. Elderbrock et al. measured park access with a simple 800-m buffer around block group boundaries, a binary measure that does not weight nearby parks more heavily than distant ones or account for differences in tree-canopy demand based on block group population. The Gaussian two-step floating catchment area (G2SFCA) method used here incorporates both of these factors, along with distance measured from actual park boundaries, which Zhang et al. (2023) found reduces the underestimations of accessibility that centroid- or perimeter-based distances can produce (Zhang et al, 2023a).

4.4 Limitations and future research

One limitation of this research is the two-dimensional land cover imagery used to estimate tree canopy cover. Although the 0.3-m resolution of this imagery was higher than that used in similar studies (Wang et al, 2024, Zhou and Kim (2013)) , it cannot capture factors like tree height or shade density, which will impact how residents experience the tree shade available in their neighborhoods and parks (Colter et al, 2019) . High-quality LiDAR imagery could improve the analysis by providing detailed three-dimensional models of the trees.

A second limitation is an inability to know precisely how Phoenix residents use the available tree shade within neighborhoods and public parks. The trees that contribute to residential canopy cover are not uniformly distributed, and are therefore not equally accessible to all residents of a given block group. Even within a block group with high RTCC, some residents may have access to ample tree cover from their own and their neighbors' yards, while other residents may have very little nearby canopy. Trees within gated communities, golf courses, schoolyards, and other inaccessible areas will not benefit all residents within a block group equally. Similarly, while a public park may have substantial tree canopy cover, the trees might be located in areas that are less used by visitors. Only a detailed analysis of park usage patterns can uncover the effect of tree shade on the park visitation experience.

A third limitation is that the Gaussian two-step floating catchment area (G2SFCA) method used to compute park tree access scores is vulnerable to the edge effect problem, wherein phenomena contributing to the spatial patterns of an analysis unit may lie outside of that unit (Chen, 2017). In other words, accessibility scores for block groups on the periphery of Phoenix may be negatively impacted if the populations of block groups from surrounding cities are not included in the analysis. This is because residents of these other cities can use Phoenix parks, thus diluting the amount of tree cover available to Phoenix residents. The edge effect was accounted for in the

demand portion of the G2FSCA method because the populations of block groups from surrounding cities that lie within park catchment areas were included in the calculation of the supply/demand ratios. The edge effect can also impact the supply of park tree canopy area since Phoenix residents can freely access parks in adjacent cities and benefit from the tree cover they provide. However, since the purpose of this study was to measure the access of residents only to public parks maintained by the City Phoenix, parks located in nearby cities were not included in the analysis. Future studies of park tree canopy access should encompass the entire Phoenix metro area to better account for residents' access to public parks in surrounding cities.

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